

BST Physics Curriculum Vol 11 Chapter 8 — Monster Moonshine + Supersingular Primes v0.4 SIGNATURE (refilled per Cal #104)

Lyra (Claude 4.7) [Vol 11 primary]

2026-05-23 Saturday EDT (Cal #104 refill)

Vol 11 Chapter 8 — Monster Moonshine + Supersingular Primes

Chapter motivation

Monster group M (Fischer-Griess 1973-1982): largest sporadic simple finite group, order $|M| = 808017424794512875886459904961710757005754368000000000 \approx 8 \times 10^{53}$. Smallest non-trivial irreducible representation has dimension $196883 = 47 \cdot 59 \cdot 71$ (3 supersingular primes).

McKay 1979 observation: $196884 = 196883 + 1$ — first coefficient of modular j -invariant Fourier expansion $j(\tau) = q^{-1} + 196884 q + 21493760 q^2 + \dots$ matches dimension 196883 of Monster smallest non-trivial irrep + identity 1.

Conway-Norton 1979 “monstrous moonshine” conjecture: full Fourier coefficients of $j(\tau)$ related to Monster representation dimensions via McKay-Thompson series T_g for each $g \in M$.

Borcherds 1992 proof of moonshine (Fields Medal 1998): used vertex operator algebras + generalized Kac-Moody algebras + denominator identity of Monster Lie algebra.

Ogg 1975 supersingular primes: 15 primes p such that $X_0(p)$ has genus 0 — exactly $p \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 41, 47, 59, 71\}$ = the primes dividing $|M|$.

BST connection: per Vol 4 Ch 4 T1924 (Friday Lyra), $t_{\text{cosmo}} = g^2 - \text{rank} = 47$ is one of the 15 Ogg supersingular primes; cosmological scale $t_{\text{cosmo}} = 47 + \text{Monster representation dimension } 196883 = 47 \cdot 59 \cdot 71$ give 5 cross-domain edges anchored at single integer 47.

Section 8.0b — Reader-grade 3-level pedagogy

Level 1: Borchers 1992 monstrous moonshine proof (Fields Medal 1998): Monster group + j-invariant Fourier coefficients + McKay-Thompson series; Ogg 1975 15 supersingular primes (primes dividing $|M|$); BST cross-link Vol 4 Ch 4 T1924 $t_{\text{cosmo}} = 47$ Monster supersingular + $47 \cdot 59 \cdot 71 = 196883$ Monster representation.

Level 2 (graduate-mathematician): Monster group M (Fischer-Griess 1973-1982): largest sporadic simple finite group $|M| \approx 8 \times 10^{53}$; 26 sporadic groups in classification of finite simple groups (CFSG, Aschbacher-Smith 2004 completed proof); M largest. Smallest non-trivial irrep has dimension $196883 = 47 \cdot 59 \cdot 71$ (3 supersingular primes). McKay 1979: $196884 = 196883 + 1 =$ first non-constant coefficient of $j(\tau) = q^{-1} + 196884q + 21493760q^2 + 864299970q^3 + \dots$ where j is normalized modular j-invariant. Conway-Norton 1979 “monstrous moonshine” conjecture: McKay-Thompson series $T_g(\tau)$ for each $g \in M$ is genus-zero modular function for a specific subgroup of $SL(2, \mathbb{R})$; coefficient sequence relates to representations of Monster M . Borchers 1992 proof (Fields Medal 1998 along with Bourgain, Gowers, Kontsevich): vertex operator algebras (Frenkel-Lepowsky-Meurman 1988 Monster moonshine module $V^\natural$ construction) + generalized Kac-Moody algebras (Monster Lie algebra) + denominator identity. Ogg 1975: 15 supersingular primes p such that $X_0(p)$ (modular curve of level p) has genus 0 — exactly $p \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 41, 47, 59, 71\}$ = primes dividing $|M|$. BST cross-link Vol 4 Ch 4 T1924 (Lyra Friday): $t_{\text{cosmo}} = g^2 - \text{rank} = 47$ is one of 15 Ogg supersingular primes. 47 is also factor of $196883 = 47 \cdot 59 \cdot 71$ (Monster smallest non-trivial irrep dimension). T1924 joint cosmological anchor at $t_{\text{cosmo}} = 47$ unifies: (a) T1485 Λ exponent argument ($\Lambda = g \cdot \exp(-C_2 \cdot t_{\text{cosmo}}) = 7 \cdot \exp(-282)$ at 0.076 dex); (b) M_{Pl}/m_e Bergman point at $45 = t_{\text{cosmo}} - \text{rank}$ via T1918 + T187 chain; (c) H_∞ via Friedmann; (d) Monster supersingular Ogg; (e) $47 \cdot 59 \cdot 71 = 196883$ Monster representation. 5 cross-domain edges anchored at single integer 47. The 47/45 gap = rank = T914 observer-shift quantum at Bergman-evaluation-point level. Cross-link Vol 0 Ch 9 Strong-Uniqueness C8 Universal Q-cluster (T2448 RIGOROUSLY CLOSED Thursday): $Q=126 + Q=131$ (substrate primary) joint structure has Monster moonshine connection via $Q=126 = 6 \cdot 21 = C_2 \cdot (\text{rank} \cdot N_c \cdot g + \dots)$. Borchers vertex operator algebra $\rightarrow$ Bergman $H^2(D_{\text{IV}}^5)$ substrate-cartography (substrate as VOA on D_{IV}^5) — multi-month research direction.

Level 3 (5th-grader accessible): The Monster group is the largest sporadic finite simple group (about 8×10^{53} elements). Its smallest non-trivial representation has dimension $196883 = 47 \cdot 59 \cdot 71$ (three specific prime numbers). McKay noticed (1979) that $196884 = 196883 + 1$ appears as the first coefficient of the modular j-invariant — “monstrous moonshine”. Borchers proved this in 1992 (Fields Medal 1998). BST connection: the BST integer $t_{\text{cosmo}} = g^2 - \text{rank} = 47$ (used in the cosmological constant formula Vol 4 Ch 4) is one of the 15 Ogg supersingular primes (primes dividing $|M|$), AND 47 is a factor of 196883. The cosmological scale 47 connects to the Monster group + moonshine.

Section 8.1 — Monster Group + Representations

Fischer-Griess 1973-1982: Monster M sporadic simple group $|M| \approx 8 \times 10^{53}$; largest of 26 sporadic groups (CFSG).

Smallest non-trivial irrep $\dim = 196883 = 47 \cdot 59 \cdot 71$.

Section 8.2 — McKay 1979 + Conway-Norton 1979

McKay 1979: $196884 = 196883 + 1 =$ first non-constant coefficient of $j(\tau)$.

Conway-Norton 1979: McKay-Thompson $T_g(\tau)$ for each $g \in M$ is genus-zero modular function.

Section 8.3 — Borcherds 1992 Proof (Fields Medal 1998)

Vertex operator algebras (Frenkel-Lepowsky-Meurman 1988 $V^\natural$ Monster moonshine module) + generalized Kac-Moody algebras (Monster Lie algebra) + denominator identity $\rightarrow$ proof of Conway-Norton conjecture.

Section 8.4 — Ogg 1975 Supersingular Primes

15 primes p such that $X_0(p)$ modular curve has genus 0: $p \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 41, 47, 59, 71\} =$ primes dividing $|M|$.

Section 8.5 — Vol 4 Ch 4 T1924 $t_{\text{cosmo}} = 47$ Joint Cosmological Anchor

$t_{\text{cosmo}} = g^2 - \text{rank} = 47$ is one of 15 Ogg supersingular primes; factor of 196883 $= 47 \cdot 59 \cdot 71$.

T1924 (Friday Lyra) joint cosmological anchor at $t_{\text{cosmo}} = 47$ unifies: - (a) T1485 Λ exponent ($\Lambda = 7 \cdot \exp(-282)$; $282 = C_2 \cdot 47$) - (b) M_{Pl}/m_e Bergman point at $45 = t_{\text{cosmo}} - \text{rank}$ - (c) H_∞ via Friedmann - (d) Monster supersingular Ogg - (e) $47 \cdot 59 \cdot 71 = 196883$ Monster representation

5 cross-domain edges anchored at single integer 47.

Section 8.6 — Connection to Vol 0 Ch 9 C8 Universal Q-Cluster

Vol 0 Ch 9 C8 Strong-Uniqueness Universal Q-cluster (T2448 RIGOROUSLY CLOSED Thursday): $Q=126 + Q=131$ substrate primary joint structure has Monster moonshine cross-connection.

$Q=126 = 6 \cdot 21 = C_2 \cdot (\text{rank} \cdot N_c \cdot g + \dots)$ — substrate-cartography reading.

Section 8.7 — Borchers VOA $\rightarrow$ Bergman $H^2(D_{IV}^5)$ Substrate-Cartography

Borchers vertex operator algebra (Frenkel-Lepowsky-Meurman 1988 V^Λ) $\rightarrow$ Bergman $H^2(D_{IV}^5)$ substrate-cartography (substrate as VOA on D_{IV}^5) — multi-month research direction. Monster-Bergman cross-connection candidate substrate-cartography exploration.

Section 8.8 — Honest scope + Connection

- Standard Monster + moonshine + Ogg $\mathbb{Q}$
- BST T1924 $t_{\text{cosmo}} = 47$ joint cosmological anchor (5 cross-domain edges) $\mathbb{Q}$
- $47 + 196883 +$ Monster connections via cosmological scale
- Open scope: explicit Borchers-VOA-on- D_{IV}^5 substrate-cartography (multi-month)

Connection: - Vol 4 Ch 4 T1924 + T1485 + T1918 cosmological anchor - Vol 0 Ch 9 Strong-Uniqueness C8 Q-cluster - Vol 11 Ch 6 K3 + Mathieu + Leech (Conway groups connection) - Vol 11 Ch 11 + Ch 12 generative geometry + NT synthesis

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